Arrival · Supported

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Pair

What does heterozygous Aa mean?

Help: Read or point first. Compare the two alleles.

2. Gamete

How many alleles for this gene go in one gamete?

Help: Read or point first. One parent contributes one to the offspring pair.

3. Options

Which gamete alleles can an Aa parent produce?

Help: Read or point first. Separate the pair, not copy Aa into every margin.

4. Combine

A gamete with A combines with a gamete with a. What is the offspring genotype?

Help: Read or point first. Use one allele from each.

My response: _____

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Version _____

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Independent · Supported

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Construct

Complete Aa × Aa with labelled gametes and four offspring boxes.

Help: Write parents first. Put one allele in each margin. Fill each box from its column and row.

2. Compare

Construct Aa × aa and state the chance of aa.

Help: Count matching boxes ___ out of four. Divide by four and multiply by 100 to give the percentage.

3. Explain

Why do two boxes sometimes contain the same genotype?

Help: Trace the column and row for each matching box. Different gamete combinations can give the same allele pair.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Supported

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

1. Construct

For Aa × Aa, list the four predicted genotype boxes.

Help: Use: my answer is ___ because ___. Point to a model or card first. Combine the row and column labels.

2. Interpret

What fraction of boxes is heterozygous?

Help: Use: my answer is ___ because ___. Point to a model or card first. Count Aa, including the equivalent aA combination.

My response: _____

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Arrival · Independent

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Pair

What does heterozygous Aa mean?

Help: Compare the two alleles.

2. Gamete

How many alleles for this gene go in one gamete?

Help: One parent contributes one to the offspring pair.

3. Options

Which gamete alleles can an Aa parent produce?

Help: Separate the pair, not copy Aa into every margin.

4. Combine

A gamete with A combines with a gamete with a. What is the offspring genotype?

Help: Use one allele from each.

My response: _____

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Independent · Independent

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Construct

Complete Aa × Aa with labelled gametes and four offspring boxes.

Help: Combine one letter from the top and one from the side.

2. Compare

Construct Aa × aa and state the chance of aa.

Help: The aa parent contributes only a.

3. Explain

Why do two boxes sometimes contain the same genotype?

Help: Different gamete combinations can give the same pair.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Independent

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

1. Construct

For Aa × Aa, list the four predicted genotype boxes.

Help: Combine the row and column labels.

2. Interpret

What fraction of boxes is heterozygous?

Help: Count Aa, including the equivalent aA combination.

My response: _____

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Arrival · Aspire

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Pair

What does heterozygous Aa mean?

Help: Compare the two alleles.

2. Gamete

How many alleles for this gene go in one gamete?

Help: One parent contributes one to the offspring pair.

3. Options

Which gamete alleles can an Aa parent produce?

Help: Separate the pair, not copy Aa into every margin.

4. Combine

A gamete with A combines with a gamete with a. What is the offspring genotype? Give a reason.

Help: Use one allele from each.

My response: _____

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Independent · Aspire

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Construct

Complete Aa × Aa with labelled gametes and four offspring boxes.

Help: Combine one letter from the top and one from the side.

2. Compare

Construct Aa × aa and state the chance of aa.

Help: The aa parent contributes only a.

3. Critique a square

A pupil places AA and aa in the margins of Aa × Aa. Explain and repair the error.

Help: The margins contain gametes, not diploid offspring.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Aspire

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

1. Construct

For Aa × Aa, list the four predicted genotype boxes.

Help: Combine the row and column labels.

2. Critique a square

A pupil places AA and aa in the margins of Aa × Aa. Explain and repair the error.

Help: The margins contain gametes, not diploid offspring.

My response: _____

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Word page

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Word	Meaning
Allele	A version of a gene.
Genotype	The allele pair for a gene.
Phenotype	An observable trait.
Dominant	Shown with one copy in this simple model.
Recessive	Shown with two recessive copies in this model.
Gamete	A sex cell with one allele for each gene in this model.
Probability	The chance of an outcome.
Expected	What the model predicts on average.
Observed	What was counted in a sample.

My language: _____

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Knowledge organiser

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Construct a monohybrid Punnett square and explain how gametes combine to predict offspring genotypes.

Aa × Aa · side / top	**A**	**a**
A	AA	Aa
a	Aa	aa

AA : Aa : aa is 1 : 2 : 1. With complete dominance, the phenotype ratio is 3 : 1. The aa chance is 1/4 = 25%.

Put one allele in each margin. Use one from the top and one from the side in each box.

This is a simple model for one gene. A dominant allele is not stronger, better or more common. Each fertilisation is a new chance. Do not use real family traits or ask for personal or medical details.

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Starter

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

What belongs outside a Punnett square?

A. One gamete allele in each margin position

B. The full parent genotype Aa in every margin

C. The number of babies in the family

Think first. Choose and give one science reason.

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Describe the parents

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Parent 1	Parent 2
Aa	Aa

For this one-gene, complete-dominance model, both parents are heterozygous: Aa × Aa.

This is a simple model for one gene. Choose when to move to the next step.

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Separate into gametes

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Each parent can produce gametes carrying A or a. Put one allele in each margin box.

This is a simple model for one gene. Choose when to move to the next step.

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Combine one from each

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Aa × Aa · side / top	**A**	**a**
A	AA	Aa
a	Aa	aa

Each offspring box combines one allele from the top with one from the side: AA, Aa, Aa and aa.

This is a simple model for one gene. Choose when to move to the next step.

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Read the probability

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Aa × Aa · side / top	**A**	**a**
A	AA	Aa
a	Aa	aa

The chance of aa is one out of four, or 25%, for each fertilisation. Four offspring do not have to follow this pattern.

This is a simple model for one gene. Choose when to move to the next step.

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We do · sort using the evidence

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Choose a group for each card. Give the clue for one choice. You can point or direct an adult.

Card	Your group
A	_____
a	_____
AA	_____
Aa	_____
aa	_____
Aa in one gamete box	_____

Groups: Gamete / Offspring pair / Wrong notation.

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

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I do 2 · worked example

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Construct BB × bb in a simple one-gene model.

A BB parent can contribute only B gametes.

A bb parent can contribute only b gametes.

Combine one from each: every predicted offspring is Bb.

The predicted chance of Bb is 100% under these model assumptions.

BB × bb · side / top	**B**	**B**
b	Bb	Bb
b	Bb	Bb

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Build the cross, then check the chance

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Complete the four boxes for Aa × Aa. Use one allele from each margin. Then change the second parent to aa and compare the two crosses.

Cross	Parent 1 gametes	Parent 2 gametes
Aa × Aa	A or a	A or a
Aa × aa	A or a	a only

My evidence and reasoning

One limit or useful check

My response: _____

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.

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Punnett square · working sheet

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Record the parent alleles. Fill all four boxes. Then find the chance of aa.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Compare with Aa × aa.

Aa × aa · side / top	**A**	**a**
a	_____	_____
a	_____	_____

Aa × Aa chance of aa: ____ . Aa × aa chance of aa: ____ .

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Evidence cards

Put one allele in each gamete margin.

Combine one allele from each parent in every box.

Read a genotype probability from the completed square.

Cut apart, point, or direct an adult. Match a card and its clue. Say why.

Card	Clue
A	One allele can be carried by one gamete.
a	This is the alternative gamete allele.
AA	Two A gametes can combine.
Aa	One A and one a combine.
aa	Two a gametes can combine.
Aa in one gamete box	A gamete has one allele for this gene, not the pair.

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